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VIBRATION DAMPING APPARATUS FOR ANTENNA DEVICE

Abstract

A vibration damping apparatus for an antenna device is disclosed. According to one embodiment of the present disclosure, the provided vibration damping apparatus comprises: a first fixing member having a predetermined thickness, and having at least a portion recessed to form a first recessed area; a second fixing member, which has a predetermined thickness, can be coupled to the first fixing member, and has at least a portion recessed to form a second recessed area; and at least one elastic member disposed between the first fixing member and the second fixing member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of International Application No. PCT/KR2023/017475, filed Nov. 3, 2023, which claims the benefit of Korean Patent Application No. 10-2022-0146667, filed Nov. 7, 2022 in the Korean Intellectual Property Office, the disclosures of which are incorporated herein in their entirety by reference.

TECHNICAL FIELD
[0002] The present disclosure relates to a vibration damping apparatus for an antenna device. More particularly, it relates to a vibration damping apparatus for an antenna device for seismic design of an antenna structure.
BACKGROUND ART
[0003] The description described in this part merely provides background information for the present disclosure, but does not constitute the related art.
[0004] An antenna for wireless communication is often installed on a rooftop of a building or the like in order to improve the quality of wireless signal transmission and reception. In this case, the antenna structure is directly exposed to climate and weather, and thus measures are taken to protect the antenna structure from various external factors.
[0005] Meanwhile, in recent years, abnormal weather due to global warming and the like has become more frequent around the world, and natural disasters such as earthquakes and typhoons not only cause casualties but also have a significant impact on the performance of various information and communication facilities.
[0006] On the roof of a building or the like, the antenna device is installed and fixed on a rigid support fixture, and various reinforcing structures are additionally installed to protect the antenna structure from the above-described external factors. In particular, vibrations caused by earthquakes, typhoons, and the like cause a decrease in the transmission/reception quality of wireless signals and the lifetime of the antenna device, and thus a solution for damping such vibrations is required.
[0007] However, most antenna structures use a method of increasing the rigidity of the structure itself or adding a separate heavy reinforcing structure for vibration damping, which causes a problem in that the weight of the entire structure is excessively increased and the cost is also increased.
DETAILED DESCRIPTION OF THE INVENTION
Technical Problem
[0008] Accordingly, the present disclosure is intended to solve these problems, and has a main object to provide a vibration damping apparatus for an antenna device, which may improve the anti-seismic performance of an antenna structure without excessive weight increase.
Technical Solution
[0009] According to one embodiment of the present disclosure to achieve this objective, there is provided a vibration damping apparatus, including: a first fixing member having a predetermined thickness and having at least portion recessed to form a first recessed area; a second fixing member which has a predetermined thickness, coupled to the first fixing member, and has at least a portion recessed to form a second recessed area; and at least one elastic member disposed between the first fixing member and the second fixing member.
[0010] Further, there is provided an antenna device including at least one vibration damping

apparatus, the antenna device includes a support fixture coupled to a side of the at least one vibration damping apparatus; and an antenna module coupled to the other side of the at least one vibration damping apparatus.

Effects of the Invention

[0011] As described above, according to the present embodiment, there is an effect that the anti-seismic performance of an antenna structure may be improved without an excessive increase in weight.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. **1** is a coupling diagram of a vibration damping apparatus according to an embodiment of the present disclosure.

[0013] FIG. **2** is an exploded perspective view of a vibration damping apparatus according to an embodiment of the present disclosure.

[0014] FIG. **3** is a top view of a vibration damping apparatus according to an embodiment of the present disclosure.

[0015] FIG. **4** is a diagram illustrating a part of an antenna device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0016] Hereinafter, some embodiments of the present disclosure will be described in detail with reference to exemplary drawings. Note that when components in each drawing are denoted by reference numerals, the same components are denoted by the same numerals as much as possible even if they are denoted on different drawings. In addition, in describing the present disclosure, if it is determined that a specific description of a related known configuration or function may obscure the gist of the present disclosure, the detailed description thereof will be omitted.

[0017] In describing components of embodiments of the present disclosure, reference numerals such as first, second, i), ii), a), and b) may be used. These symbols are only used to distinguish the components from other components, and the nature, sequence, order, or the like of the components is not limited by the symbols. In the specification, when a part “includes” or “comprises” an element, unless there is an explicit description to the contrary, the part may further include other elements rather than excluding the other elements.

[0018] FIG. **1** is a coupling diagram of a vibration damping apparatus according to an embodiment of the present disclosure.

[0019] FIG. **2** is an exploded perspective view of a vibration damping apparatus according to an embodiment of the present disclosure.

[0020] FIG. **3** is a top view of a vibration damping apparatus according to an embodiment of the present disclosure.

[0021] Referring to FIG. **1** to FIG. **3**, a vibration damping apparatus **10** according to an embodiment of the present disclosure includes all or a part of a first fixing member **100**, a second fixing member **200**, and at least one elastic member **300**, **320**, **340**.

[0022] The first fixing member **100** may include a first fixing member having a predetermined thickness and having at least portion recessed to form a first recessed area **120**. Here, the predetermined thickness may mean a thickness sufficient to withstand a load of an antenna module (**450** in FIG. **4**) to be described later.

[0023] The second fixing member **200** has a predetermined thickness, may be coupled to the first fixing member, and has at least a portion recessed to form a second recessed area **220**. Here, the thickness of the second fixing member **200** is preferably configured to be the same as the thickness of the first fixing member **100**, but is not necessarily limited thereto.

[0024] In FIG. **2**, the first recessed area **120** and the second recessed area **220** are recessed to form a rhombus shape in cross section, but are not necessarily limited to such a shape. Further, the first recessed area **120** and the second recessed area **220** are preferably configured to have the same shape, but are not necessarily limited thereto.
[0025] The first recessed area **120** may include a first recessed floor surface **122** formed at one end in the recessed direction. For example, the direction in which the first recessed area **120** is recessed may be, but is not necessarily limited to, a direction parallel to the Z axis in FIGS. **1** and **2**.
[0026] Further, the second recessed area **220** may include a second recessed floor surface **222** formed at one end in the recessed direction. For example, the direction in which the second recessed area **220** is recessed may be, but is not necessarily limited to, a direction parallel to the Z axis in FIGS. **1** and **2**.
[0027] The first fixing member **100** and the second fixing member **200** may be coupled such that the first recessed floor surface **122** and the second recessed floor surface **222** face each other. For example, the second recessed floor surface **222** of the second fixing member **200** may be coupled to the upper portion of the first recessed floor surface **122** of the first fixing member **100** so as to face each other, and in this case, at least a part of the second fixing members **200** may be coupled in such a manner as to cover at least a portion of the first fixing members **100**.
[0028] However, the second recessed floor surface **222** does not necessarily have to be coupled to the upper portion of the first recessed floor surface **122** to face each other, and conversely, the first recessed floor surface **122** may be coupled to the top portion of the second recessed floor surface **222** to face each other.
[0029] The vibration damping apparatus **10** according to an embodiment of the present disclosure may have a multilayer structure in which at least a part of one of the first fixing member **100** and the second fixing member **200** is configured to cover at least a part thereof. Therefore, in supporting an antenna module **450** to be described later, the vibration damping apparatus **10** may prevent the antenna module **450** from sagging due to a vertical load of the antenna module **450** itself or a vertical force caused by an earthquake.
[0030] The at least one elastic member **300**, **320**, **340** may be disposed between the first fixing member **100** and the second fixing member **200**. More specifically, the at least one elastic member **300**, **320**, **340** may be disposed in the first recessed area **120** and the second recessed area **220**.
[0031] In FIG. **2**, the at least one elastic member **300**, **320**, and **340** is composed of three elastic members, but is not necessarily limited to three elastic members, and may be composed of one or more elastic members.
[0032] Therefore, in the vibration damping apparatus **10** according to an embodiment of the present disclosure, the antenna module **450** may be supported, and at the same time, the at least one elastic member **300**, **320**, **340** may serve as a damper, so that a separate reinforcing structure for seismic design of the antenna device may not be required.
[0033] Meanwhile, the first fixing member **100** and the second fixing member **200** may be coupled by screws **150**. In this case, the screw **150** may pass through the centers of the first recessed floor surface **122** and the second recessed floor surface **222**, so that the first fixing member **100** and the second fixing member **200** may be engaged.
[0034] Referring back to FIG. **2**, the first fixing member **100** may include a first opening **125** penetrating the first recessed floor surface **122** in a direction parallel to the Z-axis direction, and the second fixing member **200** may include a coupling hole **225** closed on one side and extending in a direction parallel with the Z-axis.
[0035] Here, at least a part of the coupling hole **225** may be inserted into and fixed to the inside of the first opening **125**, and the screw **150** may be coupled to the coupling hole **225** to couple the first fixing member **100** and the second fixing member **200**.
[0036] However, the first opening **125** and the coupling hole **225** do not necessarily have to be included in the first fixing member **100** and the second fixing member **200**, respectively, and it is

also possible that the first fixing member **110** includes the coupling hole **225** and the second fixing members **200** include the first opening **125**.
[0037] Meanwhile, the first fixing member **100** may be symmetrically formed with respect to a first reference surface (not shown) perpendicular to the first recessed floor surface **122**, and the second fixing member **200** may be symmetrical with respect to a second reference surface (not shown) perpendicular to the second recessed floor surface **222**.
[0038] The first reference surface and the second reference surface are not shown in the drawings, but when referring to FIGS. **1** and **2**, a surface parallel to the YZ plane and passing through the center of the first recessed floor surface **122** may be the first reference surface, and a surface parallel to a YZ plane and crossing the center of the second recessed floor surface **222** may be the second reference surface.
[0039] The first reference surface and the second reference surface may be the same, wherein the at least one elastic member **300**, **320**, **340** may be configured to provide a restoring force to the first fixing member **100** and the second fixing member **200** upon a torque action about a central axis **250** perpendicular to the first recessed floor surface **122** and included in the first reference surface.
[0040] That is, the vibration damping apparatus **10** according to an embodiment of the present disclosure may be restored to the original state even when rotational torque due to an earthquake or the like is applied, so that efficient seismic design of the antenna device may be enabled.
[0041] The manner in which the at least one elastic member **300**, **320**, **340** provides the restoring force will be described later.
[0042] The first recessed area **120** may include a first recessed wall **124** formed in a direction protruding from the first recessed floor surface **122**. In this case, the first recessed wall **124** may be perpendicular to the first recessed floor surface **122**, but is not necessarily limited thereto.
[0043] The second recessed area **220** may include a second recessed wall **224** formed in a direction protruding from the second recessed floor surface **222**. In this case, the second recessed wall **224** may be perpendicular to the second recessed floor surface **222**, but is not necessarily limited thereto.
[0044] Further, the at least one elastic member **300**, **320**, **340** may include a first elastic member **300** and a second elastic member **320**.
[0045] In this case, the first elastic member **300** may have a shape corresponding to the first recessed wall **124** and may be disposed to be in contact with at least a part of the second fixing member **200** and the first recessed wall **124**. The second elastic member **320** may have a shape corresponding to the second recessed wall **224** and may be disposed to be in contact with at least a portion of the first fixing member **100** and the second recessed wall **224**.
[0046] Accordingly, the first elastic member **300** may provide a restoring force to the first fixing member **100** and the second fixing member **200** by being compressed or tensioned between the first recessed wall **124** and at least a portion of the second fixing member **200**, and the second elastic member **320** may provide a restoring force to the first fixing member **100** and the second fixing members **200** by being compressed and tensioned between the second recessed wall **224** and at least a part of the first fixing member **200**.
[0047] Since both the first elastic member **300** and the second elastic member **320** may provide a restoring force to the first fixing member **100** and the second fixing member **200**, a more effective anti-seismic design may be possible. However, at least one of the elastic members **300**, **320**, **340** does not necessarily include the first elastic member **300** and the second elastic member **320**, and may include only one of the first elastic member **300** and the second elastic members **320**.
[0048] On the other hand, the first recessed wall **124** may have a concave or convex shape centered relative to the first reference surface, and the second recessed wall **224** may have a convex or concave shape centered relative to the second reference surface. When the first reference surface and the second reference surface are the same, the first recessed wall **124** and the second recessed wall **224** may be formed symmetrically about the same reference surface.

[0049] In this case, the first elastic member **300** and the second elastic member **320** may also be symmetrically formed about the same reference surface. For example, the first elastic member **300** and the second elastic member **320** may also be configured to be concave or convex about the same reference surface. Therefore, the first elastic member **300** and the second elastic member **320** may efficiently provide a restoring force to the first fixing member **100** and the second fixing member **200** when a rotational torque is applied about the central axis **250**.
[0050] However, the first recessed wall **124** and the first elastic member **300** do not necessarily have to have a concave or convex shape centered relative to the first reference surface, and there is no limitation on the shape as long as the restoring force may be provided to the first fixing member **100** and the second fixing member **200** during the rotational torque action about the central axis **250**. The same applies to the second recessed wall **224** and the second elastic member **320**.
[0051] The at least one elastic member **300**, **320**, **340** may include a third elastic member **340** disposed between the first recessed area **120** and the second recessed area **220**. More particularly, for example, the at least one elastic member **300**, **320**, **340** may be disposed between the first recessed floor surface **122** and the second recessed floor surface **222**.
[0052] In addition, the third elastic member **340** may include at least one protruding portion **342**, **344**, **346**, **348** protruding from one surface of the third elastic member **330**.
[0053] Like the first elastic member **300** and the second elastic member **320**, the third elastic member **340** may also serve to provide a restoring force to the first fixing member **100** and the second fixing member **200**.
[0054] Further, the third elastic member **340** may include a second opening **345** penetrating the third elastic member **340** in a direction parallel to the Z-axis direction for screwing the first fixing member **100** and the second fixing member **200**. In this case, the center of the second opening **345** is preferably configured to coincide with the center of the first opening **125**.
[0055] On the other hand, in FIG. **2**, before the first fixing member **100** and the second fixing member **200** are coupled, the at least one elastic member **300**, **320**, **340** is configured to be formed in a separate configuration, but is not necessarily limited thereto.
[0056] For example, the first fixing member **100** and/or the second fixing member **200** may include a penetration injection hole **350** formed on at least one surface, and the at least one elastic member **300**, **320**, **340** may be formed by using a liquid injected through the penetration injection hole **350**.
[0057] In this case, the at least one elastic member **300**, **320**, **340** may be formed by curing the liquid phase injected into the space between the first fixing member **100** and the second fixing member **200** after the first fixing member **100** and the second fixing member **200** are coupled.
[0058] Hereinafter, a restoring force providing manner of the first elastic member **300** will be described with reference to FIGS. **1** and **3**. The restoring force providing manner of the first elastic member **300**, which will be described later, may be similarly applied to the case of the second elastic member **320**.
[0059] When a rotational torque about the central axis **250** acts on the vibration damping apparatus **10** according to an embodiment of the present disclosure, one side of the first elastic member **300** may be compressed and the other side may be tensioned with respect to the central axis **250**. For example, in FIG. **3**, one end in the positive X-axis direction of the first elastic member **300** may be compressed, and one end in the negative X-axis direction thereof may be tensioned.
[0060] Accordingly, a restoring force for returning the first elastic member **300** to the original shape may be generated, which is a torque in a direction opposite to the rotational torque applied to the vibration damping apparatus **10** according to an embodiment of the present disclosure. The restoring force is transmitted to the first fixing member **100** and the second fixing member **200**, so that the vibration damping apparatus **10** according to an embodiment of the present disclosure may maintain its original shape and effectively damp external vibration.
[0061] Although not illustrated in the drawings, the third elastic member **340** may also provide a restoring force to the first fixing member **100** and the second fixing member **200** as a torque in a

direction opposite to a rotational torque applied to the vibration damping apparatus **10** according to an embodiment of the present disclosure.

[0062] Referring back to FIG. **2**, the third elastic member **340** may have a disc shape. Further, for example, the at least one protruding portion **342**, **344**, **346**, **348** may include at least one vertical protruding portion **342**, **344** and/or at least one horizontal protruding portion **346**, **344**.

[0063] In this case, the at least one vertical protruding portion **342**, **344** may be configured to protrude in the thickness direction from the upper surface or the lower surface of the disk-shaped third elastic member **340**, and the at least one horizontal protruding portion **346**, **348** may be configured to project in a direction perpendicular to the thickness direction from one side surface of the disk-shaped third elastic member **340**.

[0064] On the other hand, as shown in FIG. **2**, the at least one vertical protruding portion **342**, **344** and the at least one horizontal protruding portion **346**, **348** are respectively configured as two protruding portions, and the four protruding portions may be arranged at intervals of 90 degrees, but the number and intervals of the protruding portions are not necessarily limited thereto.

[0065] When a rotational torque about the central axis **250** acts on the vibration damping apparatus **10** according to an embodiment of the present disclosure, the third elastic member **340** may provide a restoring force to the first fixing member **100** and the second fixing member **200** by the at least one protruding portion **342**, **344**, **346**, **348**.

[0066] FIG. **4** is a diagram illustrating a part of an antenna device according to an embodiment of the present disclosure.

[0067] Referring to FIG. **4**, an antenna device **40** according to an embodiment of the present disclosure may include all or a part of a support fixture **400**, an antenna module **450**, and at least one vibration damping apparatus **10**.

[0068] Here, the support fixture **400** may be coupled to one side of the at least one vibration damping apparatus **10**, and the antenna module **450** may be coupled to the other side. Therefore, the antenna module **450** may be coupled to the support **400** by the at least one vibration damping apparatus **10**, so that the antenna device **40** according to an embodiment of the present disclosure may also have a vibration resistance function as described above.

[0069] In addition, the antenna device **40** according to an embodiment of the present disclosure may include two vibration damping apparatuses **10**, and the two vibration damping apparatuses **10** may be disposed at the top and the bottom of the antenna module **450**. In this case, the two vibration damping apparatuses **10** may be disposed so that the same surface faces each other in a direction parallel to the height direction of the support fixture **400**.

[0070] That is, the two vibration damping apparatus **10** may be disposed in a form in which one vibration damping apparatus is inverted upside down from the other vibration damping apparatus, thereby ensuring strong support and durability.

[0071] The foregoing descriptions are merely illustrative of the technical concept of the present embodiment, and various modifications and variations may be made by those skilled in the art without departing from the essential characteristics of the present embodiment. Therefore, the present embodiments are not intended to limit the technical concept of the present embodiments, but are intended to be illustrative, and the scope of the technical concept of this embodiment is not limited by these embodiments. The protection scope of the present embodiment is to be construed according to the following claims, and all technical ideas within the scope equivalent thereto are construed as being included in the scope of rights of the present embodiment.

DESCRIPTION OF SYMBOLS

[0072] **10** vibration damping apparatus, **40** antenna device, **100** first fixing member, **120** first recessed area, **122** first recessed floor surface, **124** first recessed wall, **125** first opening, **150** screw, **200** second fixing member, **220** second recessed area, **222** second recessed floor surface, **224** second recessed wall, **225** coupling hole, **250** central axis, **300** first elastic member, **320** second elastic member, **340** third elastic member, **342**, **344** vertical protruding portion, **345** second opening, **346**, **348** horizontal protruding portion, **350** penetration injection hole, **400** support fixture, **450** antenna module

Claims

1. A vibration damping apparatus, comprising: a first fixing member having a predetermined thickness and having at least portion recessed to form a first recessed area; a second fixing member which has a predetermined thickness, coupled to the first fixing member, and has at least a portion recessed to form a second recessed area; and at least one elastic member disposed between the first fixing member and the second fixing member.
2. The vibration damping apparatus of claim 1, wherein the first recessed area comprises a first recessed floor surface formed at one end in a recessing direction, the second recessed area comprises a second recessed floor surface formed at one end in a recessing direction, and the first fixing member and the second fixing member are coupled such that the first recessed floor surface and the second recessed floor surface face each other.
3. The vibration damping apparatus of claim 1, wherein the at least one elastic member is disposed in the first recessed area and the second recessed area.
4. The vibration damping apparatus of claim 2, wherein the first fixing member is symmetrically formed with respect to a first reference surface perpendicular to the first recessed floor surface, and the second fixing member is symmetrically formed with respect to a second reference surface perpendicular to the second recessed floor surface.
5. The vibration damping apparatus of claim 4, wherein the first reference surface and the second reference surface are the same; and the at least one elastic member is configured to provide a restoring force to the first fixing member and the second fixing member when a torque is applied about a central axis perpendicular to the first recessed floor surface and included in the first reference surface.
6. The vibration damping apparatus of claim 4, wherein the first recessed area comprises a first recessed wall formed in a direction protruding from the first recessed floor surface, and the at least one elastic member comprises a first elastic member that has a shape corresponding to the first recessed wall and is disposed to be in contact with at least a part of the second fixing member and the first recessed wall.
7. The vibration damping apparatus of claim 6, wherein the second recessed area comprises a second recessed wall formed in a direction protruding from the second recessed floor surface, and the at least one elastic member further comprises a second elastic member that has a shape corresponding to the second recessed wall and is disposed to be in contact with at least a part of the first fixing member and the second recessed wall.
8. The vibration damping apparatus of claim 7, wherein the first recessed wall has a concave or convex shape centered relative to the first reference surface, and the second recessed wall has a concave or convex shape centered relative to the second reference surface.
9. The vibration damping apparatus of claim 1, wherein the at least one elastic member comprises a third elastic member disposed between the first recessed area and the second recessed area, and the third elastic member comprises at least one protruding portion protruding from one surface of the third elastic member.
10. The vibration damping apparatus of claim 9, wherein the third elastic member has a disc shape, and the at least one protruding portion comprises: at least one vertical protruding portion configured to protrude in a thickness direction from an upper surface or a lower surface of the third elastic member; and/or at least one horizontal protruding portion configured to protrude in a direction perpendicular to the thickness direction from one side surface of the third elastic member.
11. The vibration damping apparatus of claim 1, wherein the first fixing member and/or the second fixing member comprises a penetration injection hole formed through at least one surface, and the

at least one elastic member is formed by using a liquid phase injected through the penetration injection hole.

12. The vibration damping apparatus of claim 1, wherein the first fixing member and the second fixing member are coupled by a screw passing through centers of the first recessed area and the second recessed area.

13. An antenna device comprising at least one vibration damping apparatus according to claim 1, the antenna device comprising: a support fixture coupled to a side of the at least one vibration damping apparatus; and an antenna module coupled to the other side of the at least one vibration damping apparatus.

14. The antenna device of claim 13, wherein the at least one vibration damping apparatus includes two vibration damping apparatus disposed above and below the antenna module, and the two vibration damping apparatuses are arranged so that the same surfaces face each other in a direction parallel to a height direction of the support fixture.